

Time Left

0
sec(s)

MTH634:Quiz NO.02

Quiz Start Time: 04:24 PM, 02 September 2020

Question # 3 of 10 (Start time: 04:27:20 PM, 02 September 2020)

Total Marks: 1

Let (X, τ) be a metrizable then which of the following statement is true

Select the correct option

 Reload Math Equations



(X, τ) is separable.



(X, τ) is second countable.



(X, τ) has the countable chain collection



All of them

 Saving and loading next question...



Time Left

76
sec(s)

MTH634:Quiz NO.02

Quiz Start Time: 04:24 PM, 02 September 2020

Question # 4 of 10 (Start time: 04:28:52 PM, 02 September 2020)

Total Marks: 1

Let X and Y be topological spaces. A map $f : X \rightarrow Y$ is called an Open Map if _____

Select the correct option

 Reload Math Equations

- | | |
|----------------------------------|--|
| ☒ | for every open set $U \subseteq X$, the image $f(U) \subseteq Y$ is open. |
| ☐ | for every open set $U \subseteq X$, the image $f(U) \subseteq Y$ is closed. |
| ☐ | for every closed set $U \subseteq X$, the image $f(U) \subseteq Y$ is closed. |
| ☐ | None of them |

Click to Save Answer & Move to Next Question



Time Left

79
sec(s)

MTH634:Quiz NO.02

Quiz Start Time: 04:24 PM, 02 September 2020

Question # 5 of 10 (Start time: 04:29:13 PM, 02 September 2020)

Total Marks: 1

Let X and Y be topological spaces and if the mapping $f : X \rightarrow Y$ is bicontinuous then_____

Select the correct option

 Reload Math Equations

- | | |
|----------------------------------|---------------------------------|
| ☒ | it must be bijective. |
| ☐ | it may or may not be bijective. |
| ☐ | it should not be bijective. |
| ☐ | None of them. |

Click to Save Answer & Move to Next Question



Time Left

2
sec(s)

MTH634:Quiz NO.02

Quiz Start Time: 04:24 PM, 02 September 2020

Question # 6 of 10 (Start time: 04:29:26 PM, 02 September 2020)

Total Marks: 1

Which of the following statement is Not correct?

Select the correct option

- | | | |
|----------------------------------|---|---|
| ☐ | Cardinality of set X^X is a topological property. | / |
| ☐ | Length is not a topological property. | / |
| ☐ | Area is not a topological property. | / |
| ☒ | Boundedness is a topological property. | / |



Click to Save Answer & Move to Next Question

Time Left

79

sec(s)

MTH634:Quiz NO.02

Quiz Start Time: 04:24 PM, 02 September 2020

Question # 7 of 10 (Start time: 04:30:58 PM, 02 September 2020)

Total Marks: 1

Homomorphism of topological spaces is an equivalence relation.

Select the correct option

 Reload Math Equations

True



False



Click to Save Answer & Move to Next Question

Time Left

3
sec(s)

MTH634:Quiz NO.02

Quiz Start Time: 04:24 PM, 02 September 2020

Question # 8 of 10 (Start time: 04:31:13 PM, 02 September 2020)

Total Marks: 1

If (X, d) is a compact metric space, then _____

Select the correct option

 Reload Math Equations



every sequence is a Cauchy sequence.



every Cauchy sequence converges.



every sequence has a subsequence which is a Cauchy sequence.



every sequence which has a convergent subsequence is a Cauchy sequence.



Click to Save Answer & Move to Next Question



Time Left

2
sec(s)

MTH634:Quiz NO.02

Quiz Start Time: 04:24 PM, 02 September 2020

Question # 9 of 10 (Start time: 04:32:44 PM, 02 September 2020)

Total Marks: 1

Let $X = \{1, 2, 3, 4\}$ and $\tau = \{\emptyset, \{1, 2\}, \{3, 4\}, X\}$ be a topology on X and $A = \{2, 3\}$ is a dense set, then which of the following is true?

Select the correct option

 Reload Math Equations



(X, τ) must be a separable topology.



(X, τ) may or may not be a separable topology.



$\{1, 2\}$ is a closed set.



None of them.



Click to Save Answer & Move to Next Question



Time Left

41

sec(s)

MTH634:Quiz NO.02

Quiz Start Time: 04:24 PM, 02 September 2020

Question # 10 of 10 (Start time: 04:34:16 PM, 02 September 2020)

Total Marks: 1

A homoemorphic map between two topological spaces is always

Select the correct option

- | | | |
|----------------------------------|----------------------|---|
| ☐ | discontinuous | / |
| ☒ | continuous | / |
| ☐ | piecewise continuous | / |
| ☐ | an identity map | / |



Click to Save Answer & Move to Next Question

Time Left

68
sec(s)

MTH634:Quiz NO.02

Quiz Start Time: 04:52 PM, 02 September 2020

Question # 1 of 10 (Start time: 04:52:56 PM, 02 September 2020)

Total Marks: 1

Let $X = \{1, 2, 3, 4, 5, 6\}$ and $\tau = \{\emptyset, \{1\}, \{2\}, \{1, 2\}, X\}$ be a topology on X , then the local base (B_x) of the point $x = 3, 4, 5$ is_____

Select the correct option

 Reload Math Equations



$\{\{1\}, \{2\}, X\}$.



$\{\{2\}, \{1, 2\}, X\}$.



$\{X\}$.



None of them.

Click to Save Answer & Move to Next Question



Question # 1 of 10 (Start time: 03:05:52 PM, 29 August 2020)

Total Marks: 1

If $\vec{x}$ and $\vec{y}$ are in R^n , then _____ is called triangle inequality.

Select the correct option

[Reload Math Equations](#)☐

$$|\vec{x} + \vec{y}| \geq |\vec{x}| + |\vec{y}|$$

☐

$$|\vec{x} + \vec{y}| < |\vec{x}| + |\vec{y}|$$

☒

$$|\vec{x} + \vec{y}| \leq |\vec{x}| + |\vec{y}|$$

☐

$$|\vec{x} + \vec{y}| > |\vec{x}| + |\vec{y}|$$

[Click to Save Answer & Move to Next Question](#)

Time Left

70
sec(s)

MTH634:Quiz NO.02

Quiz Start Time: 04:52 PM, 02 September 2020

Question # 2 of 10 (Start time: 04:53:23 PM, 02 September 2020)

Total Marks: 1

Metric topology induced by $d(x, y) = \begin{cases} 0 & \text{if } x = y \\ 1 & \text{if } x \neq y \end{cases}$ is called _ _ _ _ _

Select the correct option

 Reload Math Equations

☐	usual topology
☒	discrete topology
☐	indiscrete topology
☐	None of them

Click to Save Answer & Move to Next Question



Time Left

51
sec(s)

MTH634:Quiz NO.02

Quiz Start Time: 04:52 PM, 02 September 2020

Question # 3 of 10 (Start time: 04:53:48 PM, 02 September 2020)

Total Marks: 1

If two sets A and B have same cardinality then which of the following is NOT true?

Select the correct option

 Reload Math Equations

☐

$f : A \rightarrow B$ is an invertible function.

☐

There is a one-to-one correspondence between the elements of A and B .

☒

$f : A \rightarrow B$ is not an invertible function

☐

All of them.

Click to Save Answer & Move to Next Question



Time Left

11
sec(s)

MTH634:Quiz NO.02

Quiz Start Time: 04:52 PM, 02 September 2020

Question # 4 of 10 (Start time: 04:54:31 PM, 02 September 2020)

Total Marks: 1

Let $X = \{1, 2, 3, 4\}$ and $\tau = \{\emptyset, \{1, 2\}, \{3, 4\}, X\}$ be a topology on X and $A = \{2, 3\}$ is a dense set, then which of the following is true?

Select the correct option

 Reload Math Equations



(X, τ) must be a separable topology.



(X, τ) may or may not be a separable topology.



$\{1, 2\}$ is a closed set.



None of them.



Click to Save Answer & Move to Next Question



Time Left

59

sec(s)

MTH634:Quiz NO.02

Quiz Start Time: 04:52 PM, 02 September 2020

Question # 5 of 10 (Start time: 04:55:54 PM, 02 September 2020)

Total Marks: 1

The usual metric $d(x, y) = |x - y|$ defined on-----

Select the correct option

 Reload Math Equations

☐	R
☒	R^2
☐	R^3
☐	R^n

Click to Save Answer & Move to Next Question



Time Left

65

sec(s)

MTH634:Quiz NO.02

Quiz Start Time: 04:52 PM, 02 September 2020

Question # 6 of 10 (Start time: 04:56:30 PM, 02 September 2020)

Total Marks: 1

Let X and Y be topological spaces and $f : X \rightarrow Y$ be a bijection then f is said to be homeomorphism if _____

Select the correct option

 Reload Math Equations

☒ f and its inverse must be continuous.

☐ f is continuous but its inverse not continuous.

☐ f and its inverse must be discontinuous.

☐ None of them.

Click to Save Answer & Move to Next Question

Time Left

46
sec(s)

MTH634:Quiz NO.02

Quiz Start Time: 04:52 PM, 02 September 2020

Question # 7 of 10 (Start time: 04:56:58 PM, 02 September 2020)

Total Marks: 1

Two spaces are called topologically equivalent if there exists a between two topological spaces.

Select the correct option

- | | | |
|----------------------------------|------------------|---|
| ☐ | surjective map | / |
| ☒ | homeomorphism | / |
| ☐ | an injective map | / |
| ☐ | bijection | / |

Click to Save Answer & Move to Next Question

Time Left

45

sec(s)

MTH634:Quiz NO.02

Quiz Start Time: 04:52 PM, 02 September 2020

Question # 8 of 10 (Start time: 04:57:45 PM, 02 September 2020)

Total Marks: 1

The property $d(x, y) \geq 0$ of a metric space is called-----

Select the correct option

 Reload Math Equations

☒	Reflexive property
☐	Non negativity
☐	Symmetric property
☐	Triangle property

Click to Save Answer & Move to Next Question



Time Left

68

sec(s)

MTH634:Quiz NO.02

Quiz Start Time: 04:52 PM, 02 September 2020

Question # 9 of 10 (Start time: 04:58:36 PM, 02 September 2020)

Total Marks: 1

Let X and Y be topological spaces. A map $f : X \rightarrow Y$ is called a Closed Map if _____

Select the correct option

 Reload Math Equations



for every open set $U \subseteq X$, the image $f(U) \subseteq Y$ is open.



for every open set $U \subseteq X$, the image $f(U) \subseteq Y$ is closed.



for every closed set $U \subseteq X$, the image $f(U) \subseteq Y$ is closed



None of them

Click to Save Answer & Move to Next Question



Time Left

9
sec(s)

MTH634:Quiz NO.02

Quiz Start Time: 04:52 PM, 02 September 2020

Question # 10 of 10 (Start time: 04:59:02 PM, 02 September 2020)

Total Marks: 1

Let (M, d) be a metric space then for all $a \in M$ and for all $r > 0$, $B(a, r)$ _____

Select the correct option

 Reload Math Equations

- | | |
|----------------------------------|-----------------------|
| ☒ | must be an open ball. |
| ☐ | must be a close ball. |
| ☐ | must be a close set. |
| ☐ | None of them. |



Click to Save Answer & Move to Next Question



Question # 1 of 10 (Start time: 07:18:31 PM, 14 February 2022)

Total Marks: 1

Let $X = \{1, 2, 3, 4, 5, 6\}$ and $\tau = \{\emptyset, \{1\}, \{2\}, \{1, 2\}, X\}$ be a topology on X , then the local base (B_x) of the point $x = 2$ is_____

Select the correct option

 Reload Math Equations☐ $\{\{1\}, \{2\}, X\}$.☐ $\{\{1\}, \{1, 2\}, \{2\}, X\}$.☒ $\{\{2\}, \{1, 2\}, X\}$.☐

None of them.

Question # 2 of 10 (Start time: 07:19:18 PM, 14 February 2022)

Total Marks: 1

If X has more than two points and (X, τ) be an indiscrete topology then which of the following statement is true about (X, τ) ?

Select the correct option

☒	It is Hausdorff.
☐	It is metrizable.
☐	It is not metrizable.
☐	None of them.

Question # 4 of 10 (Start time: 07:22:21 PM, 14 February 2022)

Total Marks: 1

Let $X = \{a, b, c\}$ and $\tau = \{\emptyset, \{a\}, \{a, b\}, X\}$ be a topology on X , then which of the following is NOT true ?

Select the correct option

 Reload Math Equations

All of them.

 (X, τ) be a first countable space. (X, τ) be a Topological space. (X, τ) be a second countable space.

Question # 4 of 10 (Start time: 07:22:21 PM, 14 February 2022)

Total Marks: 1

Let $X = \{a, b, c\}$ and $\tau = \{\emptyset, \{a\}, \{a, b\}, X\}$ be a topology on X , then which of the following is NOT true ?

Select the correct option

 Reload Math Equations

All of them.

 (X, τ) be a first countable space. (X, τ) be a Topological space. (X, τ) be a second countable space.

Question # 5 of 10 (Start time: 07:23:32 PM, 14 February 2022)

Total Marks: 1

Let $X = \{1, 2, 3, 4, 5, 6\}$ and $\tau = \{\emptyset, \{3\}, \{4\}, \{3, 4\}, X\}$ be a topology on X , then which of the following is true?

Select the correct option

 Reload Math Equations

- | | |
|----------------------------------|---|
| ☐ | The set $\{\emptyset, \{3\}, \{4\}\}$ is an open cover of the set $\{4\}$. |
| ☐ | The set $\{\emptyset, \{3\}, \{4\}\}$ is an open cover of the set $\{2\}$. |
| ☐ | The set $\{\emptyset, \{3\}\}$ is an open cover of the set $\{4\}$. |
| ☒ | None of them. |

Question # 6 of 10 (Start time: 07:24:21 PM, 14 February 2022)

Total Marks: 1

Which of the following statement is false?

Select the correct option

 Reload Math Equations

The set $\mathbb{R}$ with usual topology is second countable.



Any finite set with any topology is second countable.



Discrete topology on a countable set X is second countable.



Discrete topology on a real line $\mathbb{R}$ is second countable.

Question # 7 of 10 (Start time: 07:25:53 PM, 14 February 2022)

Total Marks: 1

Which of the following statement is false?

Select the correct option

 Reload Math EquationsDiscrete topology on a countable set X is second countable.The set $\mathbb{R}$ with usual topology is second countable.Discrete topology on a real line $\mathbb{R}$ is second countable

Any finite set with any topology is second countable.

Question # 7 of 10 (Start time: 07:25:53 PM, 14 February 2022)

Total Marks: 1

Which of the following statement is false?

Select the correct option

[Reload Math Equations](#)

Discrete topology on a countable set X is second countable.



The set $\mathbb{R}$ with usual topology is second countable.



Discrete topology on a real line $\mathbb{R}$ is second countable



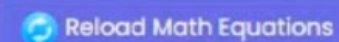
Any finite set with any topology is second countable.

Question # 8 of 10 (Start time: 07:27:24 PM, 14 February 2022)

Total Marks: 1

Let $X = \{1, 2, 3, 4\}$ and $\tau = \{\emptyset, \{1, 2\}, \{3, 4\}, X\}$ be a topology on X and $A = \{2, 3\}$ is a dense set, then which of the following is true?

Select the correct option

 Reload Math Equations

- | | |
|----------------------------------|---|
| ☐ | (X, τ) must be a separable topology. |
| ☒ | (X, τ) may or may not be a separable topology. |
| ☐ | $\{1, 2\}$ is a closed set. |
| ☐ | None of them. |

Question # 9 of 10 (Start time: 07:28:52 PM, 14 February 2022)

Total Marks: 1

Let $X = \{a, b, c\}$ and $\tau = \{\emptyset, \{a\}, \{b\}, \{a, b\}, X\}$ be a topology on X , if $B = \{\emptyset, \{a\}, \{b\}, X\}$ be the base of τ , then which of the following is true ?

Select the correct option

 Reload Math Equations

- | | |
|----------------------------------|--|
| ☐ | B be the countable base. |
| ☐ | (X, τ) be a second countable space. |
| ☒ | All of them |
| ☐ | (X, τ) be a first countable space. |

Question # 10 of 10 (Start time: 07:29:40 PM, 14 February 2022)

Total Marks: 1

Metric topology induced by $d(x, y) = |x - y|$ on $\mathbb{R}$ is called ______

Select the correct option

 Reload Math Equations

☒	discrete topology
☐	usual topology
☐	indiscrete topology
☐	None of them

Question # 1 of 10 (Start time: 03:57:43 PM, 02 September 2020)

Total Marks: 1

For a metric space (X, d) for all x, y belongs to X $d(x, y) = d(y, x)$ is called_____property.

Select the correct option

 Reload Math Equations

☐	Non negativity
☐	Reflexive
☒	Symmetric
☐	None of them

Click to Save Answer & Move to Next Question

Time Left

67

sec(s)

MTH634:Quiz NO.02

Quiz Start Time: 03:57 PM, 02 September 2020

Question # 2 of 10 (Start time: 03:57:59 PM, 02 September 2020)

Total Marks: 1

Let X and Y be topological spaces and if the mapping $f : X \rightarrow Y$ is bicontinuous then_____

Select the correct option

 Reload Math Equations

☒	it must be bijective.
☐	it may or may not be bijective.
☐	it should not be bijective.
☐	None of them.

Click to Save Answer & Move to Next Question



Time Left

18
sec(s)

MTH634:Quiz NO.02

Quiz Start Time: 03:57 PM, 02 September 2020

Question # 3 of 10 (Start time: 03:58:24 PM, 02 September 2020)

Total Marks: 1

Which of the following statement is correct?

Select the correct option

- | | | |
|----------------------------------|--|---|
| ☐ | Projection map is open. | / |
| ☐ | Projection map is closed. | / |
| ☒ | Projection map may or may not be closed. | / |
| ☐ | None of them. | / |



Click to Save Answer & Move to Next Question

Time Left

56
sec(s)

MTH634:Quiz NO.02

Quiz Start Time: 03:57 PM, 02 September 2020

Question # 4 of 10 (Start time: 03:59:39 PM, 02 September 2020)

Total Marks: 1

Let X and Y be topological spaces. A map $f : X \rightarrow Y$ is called a Closed Map if _____

Select the correct option

 Reload Math Equations



for every open set $U \subseteq X$, the image $f(U) \subseteq Y$ is open.



for every open set $U \subseteq X$, the image $f(U) \subseteq Y$ is closed.



for every closed set $U \subseteq X$, the image $f(U) \subseteq Y$ is closed



None of them

Click to Save Answer & Move to Next Question



Time Left

48

sec(s)

MTH634:Quiz NO.02

Quiz Start Time: 03:57 PM, 02 September 2020

Question # 5 of 10 (Start time: 04:00:16 PM, 02 September 2020)

Total Marks: 1

Let X and Y be topological spaces and $f : X \rightarrow Y$ be a bijection then which of the following statement is NOT true?

Select the correct option

 Reload Math Equations

- | | |
|----------------------------------|--------------------------------|
| ☐ | f is homeomorphism. |
| ☒ | f is discontinuous and open. |
| ☐ | f is continuous and closed. |
| ☐ | None of them |

Click to Save Answer & Move to Next Question



Time Left 1 sec(s)

MTH634:Quiz NO.02

Quiz Start Time: 03:57 PM, 02 September 2020

Question # 6 of 10 (Start time: 04:01:03 PM, 02 September 2020)

Total Marks: 1

Let $X = \{1, 2, 3, 4, 5, 6\}$ and $\tau = \{\emptyset, \{1\}, \{2\}, \{1, 2\}, X\}$ be a topology on X , then the local base (B_x) of the point $x = 1$ is _____

Select the correct option

 Reload Math Equations

- | | |
|----------------------------------|---|
| ☐ | $\{\{1\}, \{2\}, X\}$. |
| ☐ | $\{\{1\}, \{2\}, \{1, 2\}, X\}$. |
| ☒ | $\{\{1\}, \{1, 2\}, X\}$.  |
| ☐ | None of them |



Click to Save Answer & Move to Next Question



Time Left

4
sec(s)

MTH634:Quiz NO.02

Quiz Start Time: 03:57 PM, 02 September 2020

Question # 7 of 10 (Start time: 04:02:34 PM, 02 September 2020)

Total Marks: 1

Let $X = \{1, 2, 3, 4, 5, 6\}$ and $\tau = \{\emptyset, \{1\}, \{2\}, \{1, 2\}, X\}$ be a topology on X , then the local base (B_x) of the point $x = 2$ is _____

Select the correct option

 Reload Math Equations



$\{\{1\}, \{2\}, X\}$.



$\{\{2\}, \{1, 2\}, X\}$.



$\{\{1\}, \{1, 2\}, \{2\}, X\}$.



None of them.



Click to Save Answer & Move to Next Question



Time Left

1
sec(s)

MTH634:Quiz NO.02

Quiz Start Time: 03:57 PM, 02 September 2020

Question # 8 of 10 (Start time: 04:04:03 PM, 02 September 2020)

Total Marks: 1

Which of the following statement is Not correct?

Select the correct option

- | | | |
|----------------------------------|--|---|
| ☐ | Cardinality of set $X \times X$ is a topological property. | / |
| ☐ | Length is not a topological property. | / |
| ☐ | Area is not a topological property. | / |
| ☒ | Boundedness is a topological property. | / |



Click to Save Answer & Move to Next Question



Time Left

80

sec(s)

MTH634:Quiz NO.02

Quiz Start Time: 03:57 PM, 02 September 2020

Question # 9 of 10 (Start time: 04:05:36 PM, 02 September 2020)

Total Marks: 1

Let X and Y be topological spaces. A map $f : X \rightarrow Y$ is called an Open Map if _____

Select the correct option

 Reload Math Equations



for every open set $U \subseteq X$, the image $f(U) \subseteq Y$ is open.



for every open set $U \subseteq X$, the image $f(U) \subseteq Y$ is closed.



for every closed set $U \subseteq X$, the image $f(U) \subseteq Y$ is closed.



None of them

Click to Save Answer & Move to Next Question



Time Left

53
sec(s)

MTH634:Quiz NO.02

Quiz Start Time: 03:57 PM, 02 September 2020

Question # 10 of 10 (Start time: 04:05:49 PM, 02 September 2020)

Total Marks: 1

Every metric space is first countable.

Select the correct option

True



False



Click to Save Answer & Move to Next Question

Time Left

43
sec(s)

MTH634:Quiz NO.02

Quiz Start Time: 04:24 PM, 02 September 2020

Question # 1 of 10 (Start time: 04:24:31 PM, 02 September 2020)

Total Marks: 1

If X has more than two points and (X, τ) be an indiscrete topology then which of the following statement is true about (X, τ) ?

Select the correct option

- | | |
|----------------------------------|-----------------------|
| ☐ | It is metrizable. |
| ☐ | It is not metrizable. |
| ☒ | It is Hausdorff. |
| ☐ | None of them. |



Click to Save Answer & Move to Next Question



Time Left 15
sec(s)

MTH634:Quiz NO.02

Quiz Start Time: 04:24 PM, 02 September 2020

Question # 2 of 10 (Start time: 04:25:57 PM, 02 September 2020)

Total Marks: 1

Let $X = \mathbb{R}$ (Set of real numbers) be a usual metric space and $N \subseteq \mathbb{R}$, then which of the following is true about N ?

Select the correct option

 Reload Math Equations

- | | |
|----------------------------------|---|
| ☒ | None of its point is an interior point. |
| ☐ | It is not an open set. |
| ☐ | It has disjoint intersection with the set of its interior points. |
| ☐ | All of them. |



Click to Save Answer & Move to Next Question



Time Left 15 sec(s)

MTH634:Quiz NO.02

Quiz Start Time: 04:24 PM, 02 September 2020

Question # 2 of 10 (Start time: 04:25:57 PM, 02 September 2020)

Total Marks: 1

Let $X = \mathbb{R}$ (Set of real numbers) be a usual metric space and $N \subseteq \mathbb{R}$, then which of the following is true about N ?

Select the correct option

 Reload Math Equations

- | | |
|----------------------------------|---|
| ☒ | None of its point is an interior point. |
| ☐ | It is not an open set. |
| ☐ | It has disjoint intersection with the set of its interior points. |
| ☐ | All of them. |



Click to Save Answer & Move to Next Question

